

Script-Controlled ACL – Restrict Record Access Based on Field Value

Phase 2: Requirement Analysis

2.1 Functional Requirements

ID	Requirement	Priority
FR-1	System shall allow creation of a test user (EEE User) and custom roles bb1-bb4.	High
FR-2	System shall store student records in table u_institution_details with Branch = ECE / EEE / CSE.	High
FR-3	Users with bb1 shall read only records where Branch is EEE.	High
FR-4	Users without bb1 shall see no records.	High
FR-5	Admin shall view all records regardless of branch.	High
FR-6	Users with bb2 shall be able to create records (New button visible).	Medium
FR-7	Users with bb3 shall be able to edit records (e.g., inline list edit).	Medium
FR-8	Users with bb4 shall be able to delete records.	Medium

2.2 Non-Functional Requirements

- Security: access is denied by default unless an ACL grants it.
- Maintainability: roles and ACLs are named consistently and documented.
- Usability: restrictions apply uniformly on list views and forms.
- Reliability: behaviour must be reproducible through user impersonation.

2.3 Data Requirements - Table u_institution_details

Field	Type	Notes
Student Roll Number	Auto Number	Format INS000xxxx (e.g., INS0001007)
Student Name	Reference - User	
Faculty Name	Reference - User	
Branch	Choice	ECE, EEE, CSE - field used by the ACL
Email	String	
Phone Number	String	
Description	Multi-line String	Used to test WRITE (e.g., "ACL Is Executed")

2.4 Technical Requirements

Item	Detail
Platform	ServiceNow instance with admin access
Elevated role	security_admin (needed to create/modify ACLs)
Scripting	Server-side JavaScript in the ACL Script field (gs.hasRole)
Testing method	Impersonate user (EEE User, Admin) and open u_institution_details.list

2.5 Assumptions & Constraints

- ACLs are evaluated together: role, data condition and script must all pass for access to be granted.
- The test user is granted all four roles (bb1-bb4) for the final verification.
- The lab environment is a non-production instance.